



THE PROBLEM

Certificate fraud is a **systemic crisis** across Nigeria. Employers rely on slow, manual verification that takes **2-4 weeks** and costs up to ₦50,000 per check. Forged academic credentials pass undetected because current methods cannot detect pixel-level digital manipulation — altered names, spliced dates, and fabricated grades. Meanwhile, issuing universities have **zero visibility** into how their credentials are used and earn nothing from third-party verification.

~500K
GRADUATES
PER YEAR

68%
EMPLOYERS HIT
BY FRAUD

NO
UNIVERSITIES
EARN TODAY

THE SOLUTION — PAYMENT-GATED FORENSIC VERIFICATION



AI FORENSIC ENGINE — FOUR-LAYER ANALYSIS

VISUAL 35%

Error Level Analysis (ELA)

Re-saves as JPEG and diffs compression artifacts. Spliced names and dates produce red hotspots on the forensic heatmap.

TEXTUAL 30%

OCR + Alignment Detection

Tesseract extracts text with bounding boxes. Detects name presence, alignment anomalies, and cross-references with ELA hotspots.

NOISE 20%

Noise Consistency (Laplacian)

Laplacian filters measure local noise variance across a 4x4 grid to detect spliced regions from different sources.

EDGE 15%

Edge Coherence (Sobel)

Edge density analysis detects abrupt boundary discontinuities from cropped and pasted text blocks.

Composite Score (0-100): Four layers weighted (35/30/20/15). Score ≥ 40 = **FAIL**. Confidence via multi-layer agreement (HIGH/MEDIUM/LOW). ELA-text overlap forces minimum score of 60.

SQUAD API INTEGRATION — 4 ENDPOINTS

Initiate Payment

Collect ₦5,000 verification fee via Squad inline checkout modal
POST /transaction/initiate

Verify Transaction

Server-side confirmation before releasing forensic results
GET /transaction/verify/{ref}

Transfer API

Route ₦1,000 royalty to the issuing university's bank account
POST /payout/transfer

Sub-Merchant

Register universities as aggregator sub-merchants on Squad
POST /merchant/create-sub-users

AUTOMATED FINANCIAL PIPELINE — EVERY SCAN GENERATES REVENUE



ALIGNMENT WITH THE FOUR PILLARS



AI Automation

Automates document forensics end-to-end. ELA heatmaps replace manual visual inspection with pixel-level analysis.



Use of Data

Predictive, pixel-level data analysis detects compression signatures and alignment anomalies in real-time.



Squad APIs

4 API endpoints power the entire financial pipeline — from payment collection to automated royalty routing.



Financial Innovation

Transforms fraud prevention into a revenue stream. Universities earn passive income from every verification.